Demography, Early Voting, and Election Integrity in South Korea: Evidence Across Four Electoral Cycles

Minseong Kim

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Abstract

Allegations of fraud in South Korean elections have centered primarily on the in-precinct early-voting channel, which consistently produces higher Democratic Party vote shares than same-day ballots. We assess these claims using a unified framework applied to four elections spanning 2020–2025: the 21st General Election (2020), the 22nd General Election (2024), the 20th Presidential Election (2022), and the 21st Presidential Election (2025). First, we apply several forensic statistical tests - including simulation-adjusted second-digit Benford's law, last-digit uniformity, mixture-model fingerprinting, and evaluation of the widely-cited "63:37" early-vote ratio claim - and find no systematic pattern indicative of large-scale vote manipulation, while flagging a candidate-specific last-digit anomaly in the 2022 presidential election that warrants ongoing scrutiny. Second, we regress Democratic vote share and early-vote share on principal components derived from census, employment, and housing data at the sub-district (dong) level, with province-interacted election fixed effects and population-weighted least squares. Across all elections, sociodemographic characteristics explain 73–94% of the geographic variation in vote shares, and the same predictors perform equally well for early votes

and same-day votes. For the 2025 presidential election - a three-candidate contest - we instrument the third-party candidate's vote share using the 2017 conservative-reform candidate's support, recovering a causal estimate of vote diversion. Our findings suggest that geographic patterns in early voting are well-explained by who votes early, not by when ballots are counted.

Keywords: Election integrity; early voting; South Korea; Benford's law; sociodemographic predictors; IV-2SLS

1 Introduction

South Korea introduced nationwide in-precinct and out-of-precinct early voting in 2013, allowing any registered voter to cast a ballot at any early polling station in their registered constituency (in-precinct) or out of their constituency (out-of-precinct) during a two-day window before election day. The system has seen rapid adoption: early voting accounted for approximately 26 percent of all ballots in the 2020 general election and rose to over 37 percent by the 2025 presidential election. Alongside this growth, a distinctive and persistent empirical pattern has emerged: early ballots consistently return higher Democratic Party vote shares than same-day ballots in the same area.

This divergence has generated sustained controversy. Commentators and activists have pointed to the gap as evidence of systematic manipulation, arguing that early vote tallies are inflated for the Democratic Party through interference with the computerized vote-management system, tampering during overnight storage, or algorithmic adjustment of QR-coded ballot images. A particularly prominent variant of the allegation asserts that early votes in metropolitan Seoul, Incheon, and Gyeonggi consistently arrive at a "63:37" Democratic-to-conservative ratio regardless of constituency characteristics - a convergence alleged to be impossible under genuine random variation (Yoo and Min, 2021). These claims have circulated widely on social media, have been raised in the National Assembly, and have been the subject of multiple legal complaints. Kuk et al. (2024) show that exposure to election-fraud narratives erodes public confidence in electoral institutions even when those claims are subsequently rebutted, underscoring the importance of rigorous empirical response.

We address the fraud claims through two complementary strategies. First, we apply standard forensic statistical tests to the precinct-level ballot counts of all four major elections held between 2020 and 2025. These tests, drawn from the established election-integrity literature (Beber and Scacco, 2012; Klimek et al., 2012; Shikano and Mack, 2011; Deckert et al., 2011), detect patterns that are mechanically impossible under genuine voting - anoma-

lous digit distributions, implausibly low variance, or simultaneous spikes in both turnout and vote share. Second, we estimate population-weighted regression models that predict Democratic vote share and early-vote share from principal components of sub-district demographic, employment, and housing characteristics. A central objective is to show that the same sociodemographic factors that explain same-day vote shares predict early vote shares with equal fidelity, directly undermining the premise that early ballots follow a different generative process.

The 2025 presidential election adds an important methodological complication: the entry of Lee Joon-suk (LJS), a centrist-conservative third-party candidate, is suspected to have substantially diverted votes from all blocs. To recover a consistent estimate of Democratic Party support in 2025, we instrument LJS's vote share at the dong level using the 2017 vote share of Yoo Seung-min, a centrist-conservative presidential candidate who mentored LJS politically, and whose 2017 support base provides an arguably valid source of quasi-random variation in LJS's local appeal.

Our principal findings are threefold. Forensic tests reveal no systemic evidence of large-scale ballot manipulation across the four elections, though a candidate-specific last-digit anomaly in early Democratic votes in the 2022 presidential election merits further scrutiny and cannot be dismissed as a counting artifact. Regression analysis demonstrates that between 73 and 94 per cent of the geographic variation in Democratic vote shares - and roughly the same share of variation in early vote shares - is explained by observable demographics and labor-market structure. The 63:37/fixed ratio claim is not supported: early-voting two-party ratios vary substantially across elections and regions in a manner consistent with demographic sorting into the early-voting channel.

2 Institutional Background

2.1 The South Korean Electoral System

South Korea uses a mixed electoral system for National Assembly elections, combining single-member plurality districts with a proportional-representation tier. Presidential elections use a single nationwide plurality contest. The National Election Commission (NEC) administers elections and is formally independent of the executive.

Early voting (*sajeontupyo*) was introduced for all national elections in 2013 following a pilot in 2012 by-elections. Voters may cast their ballot at any designated early-voting station on either of the two days preceding election day. Two types of early votes are distinguished in the official records: in-precinct early votes (*gwannaesajeontupyo*), cast at the voter's registered local station, and out-of-precinct early votes (*gwanoesajeontupyo*), cast outside the voter's registered constituency and consolidated at the constituency level before proportional redistribution. This study focuses on the in-precinct channel, where geographic attribution is exact; out-of-precinct ballots are proportionally allocated across sub-districts within each constituency in proportion to each sub-district's total vote volume.

2.2 The Early-versus-Same-Day Divergence

The persistent Democratic advantage in early ballots relative to same-day ballots has multiple plausible demographic origins. Survey evidence and academic commentary (Yoo and Min, 2021) suggest that early voters skew younger, more urban, and more highly educated than same-day voters - all characteristics associated with Democratic Party support. Voters with inflexible work schedules, the elderly, and rural residents may find election-day voting more accessible or habitual. If early voters systematically differ in their partisan preferences from same-day voters, an unconditional gap between the two channels is expected under fair elections. Our regression analysis tests precisely this proposition.

2.3 The 2025 Presidential Election and the Three-Candidate Race

The 21st Presidential Election held in June 2025 was distinctive in two respects. First, it was a snap election triggered by the impeachment and removal of President Yoon Suk-yeol following his short-lived declaration of martial law in December 2024. Second, it featured a consequential third-party candidacy: Lee Joon-suk (LJS) of the Reform Party (*Gaehyukshin-dang*), running on a centrist-conservative platform distinct from the main opposition People Power Party (PPP) candidate Kim Moon-soo. This complicates direct comparison with other elections in the sample and necessitates the instrumental-variable strategy described in Section 5.

2.4 The Fraud Allegations

Fraud claims in South Korean elections have taken several forms across the period under study. The most persistent concerns the early-voting channel and includes: (i) alleged tampering during overnight storage of ballot boxes between early-voting days; (ii) alleged manipulation of QR codes embedded in early ballots during scanning; (iii) the "63:37" ratio allegation, which claims that early Democratic vote shares in metropolitan constituencies converge implausibly to an approximately fixed two-party ratio irrespective of constituency characteristics; and (iv) claims that statistical patterns in official vote counts cannot be reconciled with genuine random variation. The 63:37 figure first gained prominence following the 2020 general election and has been cited repeatedly in subsequent cycles despite varying empirical evidence.

3 Data

3.1 Unit of Analysis and Election Results

The unit of analysis in this paper is the *dong* (sub-district, or eup-myeon equivalent), the smallest administrative division at which election results are reported in South Korea. Dong-level vote counts are obtained from official records published by the National Election Commission via Korea's Public Data Portal. General election results are sourced from the 21st (2020) and 22nd (2024) National Assembly election datasets; presidential election results from the 20th (2022) and 21st (2025) presidential election datasets. After removing procedural observations (absentee, overseas, and provisional ballots), restricting only to constituencies where Democrats actually ran, and applying minimum vote-volume thresholds of fifty ballots per channel, the usable sample sizes are 3,503 dongs for the 2020 general election, 2,751 for 2024, 3,556 for the 2022 presidential election, and 3,438 for the 2025 presidential election (after merging with the 2017 Yoo Seung-min instrument). The four-election pooled panel contains 11,436 dong-election observations after merging with sociodemographic covariates.

We construct four outcome variables. *Vote share* is the Democratic Party's share of all valid ballots, combining in-precinct early, out-of-precinct early (proportionally allocated), and same-day channels. *Early democratic share* is the Democratic share computed solely over early ballots (in-precinct and allocated out-of-precinct). *Turnout* is total votes cast divided by the registered electorate, with out-of-precinct ballots and electors allocated proportionally. *Early turnout* is the share of the electorate that voted through any early channel.

3.2 Sociodemographic Covariates

Age-gender composition is derived from monthly resident registration data published by the Ministry of the Interior and Safety (MOIS). We compute, for each gender, the share of the voting-age population (18 and above) falling into seven age bands: 18–24, 25–29, 30–34,

35–39, 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, and 70-plus, yielding 22 age-gender share variables.

Employment and occupational structure is obtained from the Korean Statistical Information Service (KOSIS) at the city/county/district (*sigungu*) level, using the two semi-annual periods immediately preceding each election. Variables include gender-specific employment rates by age group, shares of wage, regular, and non-wage employment, shares of workers in six occupational categories (professional, clerical, service-sales, skilled-machine, simple labour, farming), shares of workers in six industry categories (agriculture, manufacturing, construction, retail-food, transport-finance, services), and non-participation rates by sex and age.

Apartment prices are measured as the median transaction price per square metre at the dong level, sourced from the Ministry of Land, Infrastructure and Transport's apartment transaction registry for the twelve months preceding each election, log-transformed.

Polling station density is captured through the ratio of registered voters to normal-day polling stations (*voters per station*), sourced from NEC station-location data.

3.3 Dimensionality Reduction via PCA

Because the 51 demographic and employment variables are highly collinear, we apply principal component analysis (PCA) to the combined feature set (age-gender shares, employment structure, log apartment price) separately for each isolated election and for the pooled panel. We retain components sufficient to explain 90% of the total variance, yielding 11–12 principal components per specification. Prior to regression, each component score is demeaned within province (to remove provincial-mean differences in demographic composition) and then standardized to unit variance across dongs. The interpretation of the leading components is consistent across specifications. PC1 is a *rural-elderly* axis, loading positively on the share of residents aged 60 and above, farming employment, and non-wage self-employment, and negatively on younger cohorts (20–40), wage employment, and white-collar occupations.

PC2 reflects *male formal employment intensity*, with positive loadings on male employment rates across prime and middle ages and negative loadings on the service-retail sector and older female workers. PC3 captures *industrial-manufacturing* structure, loading on skilled machine operators and manufacturing industry share. Housing wealth enters primarily through PC9 and PC12, the latter also capturing residential construction activity.

4 Forensic Statistical Tests

We apply forensic tests adapted from the election-integrity literature to ballot-count data for each election. Table 1 summarizes the results.

4.1 Digit-Based Tests

Second-digit Benford's law (F1, F7a, F7d) - in-precinct votes only. Benford's law predicts that in naturally generated numerical data spanning multiple orders of magnitude, the second significant digit follows a specific logarithmic distribution (Hill, 1995). Deckert et al. (2011) argue that 2BL tests are poorly suited to election data because precinct counts do not span the required scale range, and Mebane's (2011) rebuttal acknowledges that inference must account for data homogeneity. We therefore follow Shikano and Mack (2011) and apply a simulation-based correction (F7d) for in-precinct early votes (therefore, the results are not artifacts of weighting) in metropolitan constituencies (Seoul, Incheon, Gyeonggi), where the scale range is most restricted at the dong-aggregation level. Under this procedure, an observed chi-square statistic is compared against the 95th percentile of a null distribution generated from 500 binomial simulations calibrated to each constituency's actual vote totals and estimated Democratic share. A constituency is flagged only if its empirical chi-square exceeds the simulated threshold.

In the simulation-adjusted test, the flagging rate across constituencies is 7.9% in the 2020 general election (slightly above the 5% chance expectation), 3.4% in 2024, 1.3% in

2022, and 5.0% in 2025. Only the 2020 result marginally exceeds statistical expectation; the remaining three elections pass comfortably. The 2020 pattern was previously noted by Mebane (2020), and Park (2020) raised methodological concerns about eForensics model inputs in that election cycle. Our simulation-adjusted approach mitigates the over-rejection tendency documented by Shikano and Mack (2011) for small, homogeneous units.

Last-digit uniformity (F2) - in-precinct votes only. Genuine counting processes produce approximately uniform last-digit distributions in vote totals; human fabrication tends to over-represent round digits (Beber and Scacco, 2012). We test three parallel series: in-precinct early Democratic votes, in-precinct early conservative votes, and same-day Democratic votes. All three series pass the chi-square test ($p > 0.05$) in the 2020 general, 2024 general, and 2025 presidential elections.

The 2022 presidential election presents a notable exception. The in-precinct early Democratic vote series fails at $\chi^2 = 24.98$ ($p = 0.003$), while the in-precinct early conservative series ($\chi^2 = 9.43$, $p = 0.399$) and same-day Democratic series ($\chi^2 = 8.42$, $p = 0.492$) both pass. Moreover, the observed deviation substantially exceeds what binomial simulation of the actual vote totals under the estimated Democratic share would predict ($\chi^2_{\text{sim}} = 6.77$, $p = 0.661$). This asymmetric failure - affecting only one candidate in one channel - cannot be attributed to a counting-machine artifact that would affect all candidates equally.

We treat this finding with appropriate caution. Cramér's V for the failing series is 0.028, indicating a small though statistically significant departure from uniformity. The anomaly does not appear in the same election's conservative or same-day series, nor in any series of the other three elections. Standing alone, a small last-digit deviation does not constitute evidence of large-scale fraud: it could reflect aggregation artifacts, rounding in partially reported sub-totals, or unmodeled heaping in the underlying counting process. The weight of evidence from the remaining tests - Benford, Beber-Scacco, and Klimek - does not corroborate systematic manipulation in 2022.

Beber-Scacco tests (F8, F9) - in-precinct early Democratic votes only. The digit-preference

test (Beber and Scacco, 2012) checks whether human fabricators over-use "round" digits (0 and 5). The observed frequency of such digits is 19.5–21.1% across the four elections (expected 20.0%), and no election produces a statistically significant departure ($|z| < 1.4$ in all cases). The mean and variance of last digits likewise conform to a uniform distribution in all four elections.

4.2 Distributional Tests

The 63:37/fixed ratio claim (F7c). Table 1 reports the macro two-party Democratic share in early in-precinct votes across the metropolitan region (Seoul, Incheon, Gyeonggi). In the 2020 general election, early two-party shares are close to 63% in Seoul (63.5%), Gyeonggi (63.3%), and Incheon (63.1%). In the 2024 general election, however, the corresponding figures are 59.6%, 61.8%, and 59.0% - substantially below 63% and diverging between regions. In the 2022 presidential election, early Democratic shares fall to 53.7%, 57.0%, and 54.5%, again diverging. In the 2025 presidential election (Lee Jae-myung versus Kim Moon-soo, with LJS as a major third candidate distorting the two-party count), the two-party Democratic early share rises to 72.0%, 73.7%, and 72.2%.

Several features of these data are inconsistent with a fixed-ratio fraud hypothesis. First, the "63:37" ratio does not hold across elections; it approximately characterizes 2020 but not 2022, 2024, or 2025. If the ratio were mechanically imposed, it should be invariant across elections and perhaps across regions. Second, within each election, the micro-level standard deviation across precincts is large: 9.1 percentage points in 2020, 7.0 in 2024, 7.0 in 2022, and 6.4 in 2025. Low variance across units is the theoretical signature of a fixed-ratio imposition; by contrast, a high-variance distribution whose aggregate converges to a mean through the law of large numbers is what one would expect from genuine democratic choice. Third, the macro ratio tracks the overall Democratic performance level closely - elections in which Democrats do better nationwide (2020 landslide, 2025 snap election under post-impeachment conditions) also show higher early Democratic shares, as expected if the same underlying

partisan geography drives both.

Mixture-model fingerprinting (F10). Klimek et al. (2012) propose that fraud generates identifiable clusters combining high turnout with high vote share, reflecting ballot stuffing. We apply BIC-selected Gaussian mixture models to the joint (turnout, vote share) distribution in each election. In all four elections, the identified clusters correspond to recognizable partisan-geographic regions (Honam Democratic strongholds, Yeongnam conservative strongholds, swing metropolitan areas) and no cluster simultaneously exhibits anomalously high turnout and high vote share. All four elections pass the Klimek fraud criterion.

5 Regression Framework

5.1 Baseline WLS Specification

For each of the three elections in which the Democratic Party runs effectively in a two-candidate race (21st General 2020, 22nd General 2024, 20th Presidential 2022), we estimate:

$$Y_{d,e} = \alpha_{p(d),e} + \sum_{k=1}^{K} \beta_k \widetilde{z}_{k,d,e} + \gamma \widetilde{w}_{d,e} + \varepsilon_{d,e} \tag{1}$$

where $Y_{d,e}$ is one of four outcomes (turnout, early turnout, vote share, early democratic share) for dong d in election e; $\alpha_{p(d),e}$ is an intercept for province $p(d)$ (or an interacted province-by-election fixed effect in the pooled specification); $\widetilde{z}_{k,d,e}$ are province-demeaned (mean computed across regions within a single province) and standardized principal component scores; and $\widetilde{w}_{d,e}$ is constituency-demeaned and standardized voters-per-station (included in turnout equations only). Each observation is weighted by the dong's total registered electorate (*sum_people*). Standard errors are clustered at the constituency (*seon'geo-gu*) level.

Vote-share and early-share equations are estimated without a turnout covariate, so that the PC coefficients recover the reduced-form relationship between demographics and partisan outcomes rather than mediated effects.

5.2 Instrumental Variable Strategy for the 2025 Presidential Election

The entry of Lee Joon-suk introduces an endogeneity problem: LJS vote share is correlated with local Democratic strength or weakness. We instrument LJS's dong-level vote share (LJS_d) using Yoo Seung-min's dong-level vote share in the 2017 presidential election ($\text{Yoo}_{d,2017}$).

The relevance condition is strongly satisfied: Yoo Seung-min and Lee Joon-suk share a direct political lineage (LJS served under Yoo and was widely regarded as his political protégé), ran on overlapping centrist-conservative platforms, and drew from the same voter pool of conservative reformers and conservative-leaning centrists dissatisfied with major parties. First-stage F-statistics exceed 109 in the isolated pres21 specification and 310 in the pooled panel, far above conventional weak-instrument thresholds.

The exclusion restriction requires that 2017 Yoo vote share affects 2025 Lee Jae-myung vote share *only* through LJS's 2025 performance, conditional on province fixed effects and sociodemographic controls. This assumption is plausible for several reasons. Yoo's 2017 electorate was a centrist group motivated by dissatisfaction with the Park Geun-hye administration; its geographic distribution reflects idiosyncratic local factors in conservative opposition politics eight years earlier. Given the substantial time elapsed, the intervening elections (2020 general, 2022 presidential, 2024 general), and the very different electoral context of the post-impeachment 2025 snap election, a direct path from 2017 Yoo support to 2025 Democratic support - independently of LJS's performance - is difficult to sustain. Residual correlation with unobserved local partisan trends is further reduced by the inclusion of province fixed effects and the full demographic PC vector.

5.3 Pooled Panel Specification

In the pooled specification, we stack all four elections and replace province fixed effects with province-by-election interaction fixed effects ($\alpha_{p(d)\times e}$), absorbing all region-specific election-cycle effects including idiosyncratic third-party dynamics in Honam or Yeongnam. The LJS instrument is *activated* only for pres21 observations (active_yoo = $\text{Yoo}_{d,2017} \cdot \mathbf{1}[\text{e} = \text{pres21}]$), and LJS vote share is set to zero for non-pres21 elections. This specification pools the demographic variation across all elections to estimate common PC coefficients while allowing baseline levels to vary freely by region-cycle. Standard errors are robust heteroskedastic (HC3) for pooled WLS comparisons and robust in the IV specifications, given the large pooled sample.

6 Results

6.1 Sociodemographic Predictability of Vote Shares

Table 2 presents the principal-component coefficients and goodness-of-fit statistics for vote share and early democratic share in the three two-candidate elections. The most striking finding is the uniformly high explanatory power: the within-province demographic variation alone explains 74.7% of the geographic variation in total vote share in 2020, 82.1% in 2024, and 93.5% in the 2022 presidential election. For early democratic share, the corresponding figures are 73.4%, 82.1%, and 94.1% - essentially identical.

The near-equality of R^2 between vote share and early democratic share is itself informative. If early ballots followed a different generative process than same-day ballots - whether through demographic self-selection into the early channel, manipulation, or any other cause - one would expect either a systematically higher or lower fit for early votes relative to total votes. The absence of any such differential is consistent with early ballots being cast by the same underlying population, with the same preferences, that the demographic model is

designed to capture.

The PC1 coefficient is negative and highly significant for Democratic vote share in all elections: in dongs where the population is older, more rural, and more reliant on non-wage farming employment (high PC1), the Democratic share is lower. This gradient is stable at approximately -0.023 to -0.030 standardized units across the three elections, and is nearly identical when estimated on early democratic share (-0.025 to -0.034). The consistency of this coefficient confirms that early voters and same-day voters in any given area respond to the same underlying partisan geography.

PC2 (male employment intensity) is positively associated with Democratic support in both general elections ($\beta \approx +0.011$ to $+0.019$ for vote share) and also, more modestly, in the 2022 presidential election ($\beta = +0.006$ for vote share, statistically significant; not distinguishable from zero for early democratic share), reflecting the somewhat different class-coalition dynamics of presidential versus general-election contests.

PC3 (industrial-manufacturing) and PC5 have mixed signs across elections, consistent with the complex relationship between industrial structure and partisan alignment in Korean politics.

Apartment prices, which load most heavily on PC12, do not produce a consistent first-order effect once provincial baselines are absorbed, suggesting that housing wealth is largely collinear with the regional fixed effects.

6.2 Turnout and Early Turnout

The sociodemographic model predicts turnout less well than vote shares: R^2 ranges from 0.33 to 0.39 in isolated elections.

This is expected because aggregate turnout is sensitive to local administrative factors - polling station access, queue length, weather - that are orthogonal to demographics. The voters-per-station variable is consistently negative and significant in all turnout equations, indicating that dongs with more voters per station have somewhat lower turnout, consistent

with a friction interpretation.

Early turnout is somewhat better predicted by demographics ($R^2 = 0.41$–0.49) than total turnout, plausibly because the early-voting decision is more strongly correlated with occupational flexibility and social networks captured by employment structure.

The PC1 (rural-elderly) coefficient is consistently positive for early turnout in the general elections, reflecting that sub-districts with larger elderly populations tend to have higher early-voting uptake in those cycles, while effects are more heterogeneous in presidential elections.

6.3 The 2025 Presidential Election: IV Results

Table 3 presents the IV-2SLS estimates for the 2025 presidential election.

In areas where LJS performed better (instrumented by 2017 Yoo vote share), the Democratic vote share is substantially lower: a one-unit increase in LJS vote share reduces Democratic share by 3.05 percentage points (robust SE = 0.42) in the isolated pres21 specification. The parallel estimate for early democratic share is -1.47 (SE = 0.30), indicating that the LJS effect is somewhat attenuated in the early-voting channel - plausibly because LJS's supporters who chose early voting were disproportionately drawn from conservative voters who would not have supported Lee Jae-myung regardless.

The effect on turnout is positive and significant (1.82, SE = 0.20): dongs with stronger LJS performance exhibit higher overall turnout, consistent with LJS mobilizing voters who would otherwise have abstained rather than solely redistributing votes from either Lee Jae-myung or Kim Moon-soo.

The IV estimate for early turnout is negative in sign but does not reach conventional significance thresholds (-0.60, SE = 0.35, $p = 0.09$), indicating that the evidence for LJS supporters preferring same-day over early voting is limited in the isolated specification and absent in the pooled panel (see below).

6.4 Pooled Panel

Table 3 reports the pooled IV panel (pooling different elections in one regression) results under the preferred specification with interacted province-by-election fixed effects.

The estimated coefficient on LJS vote share is -1.44 (SE $= 0.14$) for total democratic vote share and -0.82 (SE $= 0.12$) for early democratic share. These pooled estimates are more conservative than the isolated pres21 results because the interaction FEs absorb regional election-cycle trends that the isolated specification attributes to the LJS channel. The first-stage F-statistic remains very large ($F = 311$), ruling out weak-instrument concerns. The LJS coefficient for early turnout in the pooled panel is -0.11 (SE $= 0.13$), which is indistinguishable from zero ($p = 0.39$), confirming that no robust inference about voting channel-switching behavior can be drawn from these data.

The election fixed effects (additive specification) reveal expected patterns in the pooled turnout equation: presidential elections (pres20 and pres21) have significantly higher turnout than the reference general election (21st, 2020), with coefficients of 0.106 and 0.117 respectively. The pooled PC1 coefficient on Democratic vote share is -0.032 (interacted-FE specification), essentially identical to the isolated estimates, confirming the robustness of the rural-elderly axis across different election types.

6.5 Early Democratic Share Versus Total Vote Share

A critical test of the fraud hypothesis is whether early democratic share and total vote share diverge systematically after conditioning on demographics. If early votes were inflated for the Democratic Party beyond what the underlying electorate would support, the PC coefficients on early democratic share should be larger in magnitude (more Democratic) than those on total vote share, or the residual early share should be persistently positive after accounting for demographics.

The evidence consistently contradicts this expectation. PC1 coefficients are -0.030 (vote share) and -0.034 (early share) in 2020; -0.023 and -0.028 in 2024; and -0.024 and -0.025

in 2022. The small and directionally inconsistent differences between these pairs are well within sampling uncertainty. The intercepts confirm that early votes are somewhat more Democratic (intercepts approximately 4–7 percentage points higher for early share), but this baseline difference is absorbed by the province fixed effects and reflects composition of the early electorate, not an anomalous level shift.

7 Discussion

7.1 Implications for the Fraud Hypothesis

Taken together, our findings substantially undermine the empirical foundation of the major election fraud claims circulating in South Korean political discourse.

The claim that early vote distributions are "impossible" to explain by legitimate demographic processes is directly refuted by the regression evidence. A model using only census age-gender shares, employment structure, and housing prices - none of which are under any party's control in the weeks preceding an election - explains between 73 and 94 per cent of the geographic variance in early democratic share, and does so with coefficients nearly identical to those for total vote share. The residual variation, though unexplained, is unsystematically distributed and does not cluster in regions or election cycles that would be consistent with a targeted fraud strategy.

The fixed ratio claim is also not supported as a universal regularity. The approximately 63% early Democratic two-party share in metropolitan areas in 2020 reflects the combination of a Democratic landslide in that cycle (the party won roughly 60% of the national two-party vote) and the demographic composition of early voters. The same ratio does not emerge in 2022 or 2024, when Democratic performance was more modest, and in 2025 it rises far above 63% in response to the post-impeachment political environment. Furthermore, each capital metropolitan province vote share diverges from a mean value in 2022, 2024 and 2025. The spatial standard deviation across precincts within any single election is 6–13 percentage

points, ruling out any mechanism that imposes a spatially uniform result.

7.2 The 2022 Last-Digit Anomaly

The asymmetric last-digit irregularity in the 2022 presidential early Democratic vote series is the one finding that gives us pause. The failure is statistically significant, candidate-specific, and exceeds what binomial simulation predicts. We do not dismiss it. However, we also do not interpret it as strong evidence of fraud for the following reasons.

First, the size of the anomaly is small (Cramér's $V = 0.028$). Second, the remaining tests applied to the same election pass without exception, including the simulation-adjusted Benford test, the Beber-Scacco digits test, and the Klimek fingerprint. Fraud large enough to change the outcome of a highly competitive election (the 2022 margin was approximately 0.73 percentage points) would almost certainly generate detectable signals across multiple forensic dimensions simultaneously. Third, our regression analysis shows that 2022 early democratic share is explained by the demographic model to the same degree ($R^2 = 0.941$) as total vote share ($R^2 = 0.935$), with near-identical coefficients.

7.3 Limitations

Several limitations should be noted. Our regression analysis is observational and cannot rule out all confounds. In particular, we do not have a direct measure of *which* residents voted early versus on election day, meaning that the demographic predictors reflect the total registered population rather than the self-selected early electorate. If early voters are a non-representative draw from the local population along dimensions not fully captured by the PC model, the residuals will conflate that compositional effect with any genuine anomaly.

Employment and occupational data are measured at the sigungu (city-county-district) level rather than the dong level, introducing measurement imprecision in urban areas with heterogeneous sub-districts. Apartment prices are imputed using median matching where exact data are unavailable, which may reduce precision in rural areas.

The exclusion restriction for the Yoo-LJS instrument, while plausible, is not formally testable with a single instrument. Researchers with access to individual-level voter files or a second valid instrument could conduct an over-identification test.

8 Conclusion

We have examined four South Korean elections across 2020–2025 using two complementary analytical strategies. The forensic tests reveal no systematic pattern of large-scale ballot manipulation: most tests pass cleanly in all four elections, the "fixed ratio" claim does not hold across electoral cycles, and mixture-model fingerprinting identifies clusters driven by partisan geography rather than fraud.

The one noteworthy exception - an asymmetric last-digit irregularity in 2022 early Democratic vote counts - is statistically significant but small in magnitude and unaccompanied by corroborating anomalies in other dimensions; a technical or theoretical explanation cannot be ruled out.

The regression evidence is equally clear: between 73 and 94 per cent of the geographic variation in Democratic early vote share is explained by observable sociodemographic characteristics, and the same model explains total vote share with essentially identical fidelity. The claim that early ballot distributions are anomalous relative to demographic expectations is inconsistent with the data.

These findings have implications beyond South Korea. As early and absentee voting expands globally, early-versus-election-day divergences in partisan composition will become more common and more salient. Our analysis illustrates that such divergences are predictable from demographic sorting and do not constitute a prima facie case for fraud. Forensic tests, properly calibrated for the data-generating process of the election under study, remain the appropriate first tool of scrutiny - and must be accompanied by covariate-based plausibility assessment rather than applied in isolation.

Data availability

The data, codes and full forensic and regression results are available at github page and repository: `https://mkimacad.github.io/south_korean_election_forensics`.

Note that the github repository contains richer sets of regression results than reported in this paper.

Declaration of interests

The author(s) have no funding source to declare. Furthermore, there is no conflict of interests.

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Table 1: Summary of Forensic Test Results Across Four Elections

	General Elections		Presidential Elections	
Test	21st (2020)	22nd (2024)	20th (2022)	21st (2025)
F1: 2BL Nationwide				
χ^2 (df=9)	9.42	10.51	13.87	8.53
p-value	0.400	0.311	0.127	0.481
Result	Pass	Pass	Pass	Pass
F2: Last-Digit Uniformity				
Early Dem (χ^2)	6.14	11.05	**24.98**	11.09
Early Dem (p)	0.726	0.272	**0.003**	0.270
Early Con (χ^2)	9.63	6.65	9.43	10.49
Same-Day Dem (χ^2)	9.51	7.43	8.42	5.00
Result	Pass	Pass	**Fail** (Dem only)	Pass
F7c: Early Two-Party Dem Share				
Seoul combined	63.5%	59.6%	53.7%	72.0%
Gyeonggi	63.3%	61.8%	57.0%	73.7%
Incheon	63.1%	59.0%	54.5%	72.2%
Precinct micro std	9.1 pp	7.0 pp	7.0 pp	6.4 pp
63:37/fixed ratio claim consistent?	Partly	No	No	No
F7d: Sim-Adjusted 2BL (S&M)				
Constituencies flagged	9/114	4/119	1/77	4/80
Flagging rate	7.9%*	3.4%	1.3%	5.0%
Result	Fail	Pass	Pass	Pass
F8: Beber-Scacco 0s & 5s				
Result	Pass	Pass	Pass	Pass
F9: Last-Digit Mean & Variance				
Result	Pass	Pass	Pass	Pass
F10: Klimek Fingerprint				
No. BIC components	4	5	4	4
Fraud cluster present?	No	No	No	No
Result	Pass	Pass	Pass	Pass

Notes: For F7d, the expected flagging rate under the null is approximately 5%. An asterisk (*) denotes a rate exceeding expectation. The F2 failure for the 20th Presidential Election is asymmetric: only the early Democratic series fails; the early conservative and same-day Democratic series pass. Simulation-adjusted 2BL follows Shikano and Mack (2011) with 500 simulations and $df_t = 5$. Fingerprint analysis follows Klimek et al. (2012); all clusters are explained by regional partisan geography (Honam and Yeongnam regions).

Table 2: WLS Regressions: Democratic Vote Share and Early Democratic Share (Isolated Elections)

	21st General (2020)		22nd General (2024)		20th Presidential (2022)	
	Vote share	Early share	Vote share	Early share	Vote share	Early share
PC1 (rural-elderly)	-0.030^{***}	-0.034^{***}	-0.023^{***}	-0.028^{***}	-0.024^{***}	-0.025^{***}
	(0.006)	(0.006)	(0.005)	(0.005)	(0.003)	(0.003)
PC2 (male employment)	0.019^{***}	0.016^{***}	0.011^{***}	0.009^{**}	0.006^{***}	0.003
	(0.005)	(0.005)	(0.004)	(0.003)	(0.002)	(0.002)
PC3 (manufacturing)	-0.002	-0.002	0.005	0.004	0.007^{***}	0.006^{***}
	(0.005)	(0.005)	(0.004)	(0.003)	(0.002)	(0.002)
PC5	-0.007	-0.005	0.010^{***}	0.007^{***}	0.004^{**}	0.002
	(0.004)	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.747	0.734	0.821	0.821	0.935	0.941
N	3,503	3,503	2,751	2,751	3,556	3,556

Notes: Population-weighted least squares; weights = registered electorate (*sum_people*). All PC scores are province-demeaned and standardised. Province fixed effects use Busan as the omitted category. Standard errors (in parentheses) are clustered at the constituency level. Only selected PCs are shown; full results are available upon request - also available at the github repository. ** $p < 0.05$; *** $p < 0.01$.

Table 3: IV-2SLS Results: LJS Vote Share Instrument (21st Presidential Election Isolated and Pooled Panel)

	Turnout	Early Turnout	(Dem) Vote Share	Early Dem Share
Panel A: Isolated 21st Presidential (2025) - Province FEs, IV: Yoo 2017				
LJS vote share	1.823^{***}	-0.597	-3.052^{***}	-1.472^{***}
	(0.196)	(0.353)	(0.415)	(0.304)
First-stage F	109.34	109.34	109.50	109.50
R^2	0.356	0.649	0.896	0.923
N	3,438	3,438	3,438	3,438
Panel B: Pooled Panel - Province × Election FEs, IV: Active Yoo				
LJS vote share	0.760^{***}	-0.110	-1.436^{***}	-0.821^{***}
	(0.074)	(0.128)	(0.145)	(0.117)
First-stage F	310.96	310.96	311.04	311.04
R^2	0.752	0.628	0.876	0.883
N	11,436	11,436	11,436	11,436

Notes: Weights = registered electorate. Panel A estimates use province fixed effects and control for the full set of province-demeaned PC scores and voters-per-station (turnout equations). Panel B uses interacted province-by-election fixed effects (preferred specification). The instrument is Yoo Seung-min's dong-level vote share in the 2017 presidential election (*active_yoo*), set to zero for non-pres21 observations in Panel B. Robust standard errors in parentheses. All PC coefficients are included but not shown; turnout equations additionally include voters-per-station as a control. The Panel A R^2 for turnout is reported from the IV regression (negative values are possible when the IV model fits worse than the unconditional mean). *** $p < 0.01$.